Diabetes Risk Analysis and Prediction Using Machine Learning

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***Abstract*—This study examines whether routinely collected, self-reported health indicators can support the identification of adults reporting diabetes. We analyse the 2015 BRFSS Diabetes Health Indicators dataset, comprising 253,680 records, 21 input variables, and three outcome categories: no diabetes, prediabetes, and diabetes. The initial audit found no missing values and 23,899 exact duplicate records. We examine associations with blood pressure, body mass index (BMI), age, income, education, and perceived general health, then compare Logistic Regression, Decision Tree, Random Forest, and k-Nearest Neighbours using a stratified 80:20 train–test partition. The conventional models achieved accuracy between 83.2% and 84.6%, close to the 84.2% majority-class baseline. Their macro-F1 scores ranged from 0.392 to 0.402, while detection of prediabetes remained negligible. Applying class-balanced sampling weights to Random Forest increased macro-F1 to 0.443 and diabetes recall from 19.7% to 66.6%, although accuracy decreased by 7.3 percentage points. The deduplicated experiment produced a comparable pattern. Logistic Regression obtained the highest one-vs-rest macro AUC of 0.781. These findings show that accuracy alone is inadequate for this imbalanced task; class-sensitive measures, particularly recall and macro-F1, are more informative when selecting a screening model.**

***Index Terms*—diabetes prediction, BRFSS, class imbalance, random forest, macro F1-score, ROC-AUC**

I. Introduction

Diabetes is a major public-health concern, and a substantial number of affected individuals may remain undiagnosed for extended periods [1]. Delayed identification can postpone treatment and increase the likelihood of cardiovascular, renal, and neurological complications [2]. Laboratory testing remains central to diagnosis, yet applying it universally can be costly. A preliminary risk-assessment approach based on information people can report themselves may therefore help prioritise individuals for further clinical evaluation.

Population health surveys offer a useful basis for preliminary risk modelling because they contain demographic, behavioural, and health-related information at scale. However, responses may contain recall error and reporting bias, and the distribution is dominated by respondents without diabetes. Under this degree of class imbalance, a high overall accuracy can coexist with poor identification of minority cases. Evaluation must consequently consider how well each class is recognised rather than relying on accuracy alone.

The analysis uses the Diabetes Health Indicators extract associated with the 2015 Behavioral Risk Factor Surveillance System (BRFSS) [7], [8]. The work makes four contributions:

- a check of the file's size, missing values, duplicates, and class balance, recomputed from the CSV itself;
- an exploratory study of how diabetes relates to blood pressure, cholesterol, BMI, age, income, education, and general health;
- a side-by-side test of four classifiers, judged by per-class recall, macro-F1, confusion matrices, and multiclass ROC-AUC, repeated with and without duplicate rows;
- a small experiment on class weighting, and a frank list of what we have not tested.

II. Related Work

Prior work has applied a range of supervised learning techniques to diabetes-related prediction. In their review, Kavakiotis *et al.* identify methods including support vector machines, ensemble trees, and logistic regression as frequently used approaches [3]. Sisodia and Sisodia evaluate several classifiers using the Pima Indians dataset [4], whereas Zou *et al.* investigate decision trees, random forests, neural networks, and feature selection with clinical examination data [5]. Xie *et al.* provide a particularly relevant example by developing type 2 diabetes risk models from BRFSS survey information [6].

Our evaluation is also informed by research on imbalanced classification and model assessment. He and Garcia discuss resampling and cost-sensitive learning as strategies for handling unequal class frequencies [13]; SMOTE is one established method for generating synthetic examples of under-represented classes [14]. For evaluation, Sokolova and Lapalme explain why accuracy and F-measure-based summaries can lead to different conclusions [15]. ROC analysis and its multiclass adaptation are described by Fawcett and by Hand and Till [16],

TABLE I
PREDICTOR VARIABLES IN THE DATASET

Type	Variables (coding)
Binary (0/1)	HighBP, HighChol, CholCheck, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity, Fruits, Veggies, HvyAlcoholConsump, AnyHealthcare, NoDocbcCost, DiffWalk, Sex
Ordinal	GenHlth (1–5, excellent to poor), Age (1–13 five-year bands, 1 = 18–24, 13 = 80+), Education (1–6), Income (1–8)
Numeric	BMI (kg/m^2), MentHlth and PhysHlth (days of poor health in the past 30 days, 0–30)

TABLE II
DATASET STATISTICS VERIFIED AGAINST THE CSV FILE

Property	Value
Rows / columns	253,680 / 22
Missing cells	0
Duplicate rows (all columns identical)	23,899 (9.42%)
Rows after removing duplicates	229,781
Class 0 (no diabetes)	213,703 (84.24%)
Class 1 (prediabetes)	4,631 (1.83%)
Class 2 (diabetes)	35,346 (13.93%)
BMI minimum / median / maximum	12 / 27 / 98

[17]. In healthcare applications, the target definition itself can introduce systematic bias, as highlighted by Obermeyer *et al.* [19].

III. DATASET

A. Source and Target

The Behavioral Risk Factor Surveillance System is an annual telephone-based survey of adults in the United States, administered by the Centers for Disease Control and Prevention [7]. The file used here, `diabetes_012_health_indicators_BRFSS2015.csv`, is the cleaned 2015 dataset distributed through Kaggle [8]. Its response variable, `Diabetes_012`, encodes three groups: 0 for no diabetes (including diabetes reported only during pregnancy), 1 for prediabetes, and 2 for diabetes. Both predictor responses and outcome labels originate from self-reports.

B. Predictors

The input space contains 21 attributes, summarised in Table I. Although the file stores these fields numerically, many represent binary responses or ordered categorical levels rather than continuous measurements.

C. What We Verified

The audit results are presented in Table II. The dataset contains 253,680 observations and 22 fields, consisting of 21 predictors and one target. No cells are missing. We identified 23,899 records (9.42%) that are exact row-level duplicates. The class distribution is highly uneven: 84.24% belong to class 0, 13.93% to class 2, and 1.83% to class 1. Thus, the majority class is approximately 46 times larger than the prediabetes group.

IV. METHODOLOGY

A. Workflow

The processing sequence is outlined in Fig. 1. The CSV is imported using pandas [21], followed by data validation and exploratory analysis. We then create a stratified partition, apply scaling to the algorithms that require it, fit the classifiers through scikit-learn [20], and assess their predictions on the reserved test subset.

B. Preprocessing Choices

Missing values. The audit detected no missing entries; consequently, no imputation procedure was necessary.

Duplicates. Duplicate observations were retained for the primary experiment because identical responses to a limited set of survey questions do not necessarily indicate erroneous records. Nearly 99% of the repeated rows belong to class 0 (Section V). Nevertheless, an identical observation could occur in both the training and test partitions. We therefore conducted a second evaluation after removing duplicate rows to examine the possible influence of this overlap (Section VIII).

Scaling. Feature standardisation was applied to Logistic Regression and KNN, whose behaviour is sensitive to differences in variable magnitude. Each value was transformed as $z = (x - \mu)/\sigma$, where the mean and standard deviation were estimated exclusively from the training data. Decision trees and forests were used without scaling.

Encoding. Since all input fields are numeric in the supplied file, no categorical encoding step was added. The ordered fields Age, Education, Income, and GenHlth were retained as integer-valued variables; this representation implicitly treats successive category codes as equally spaced.

C. Stratified Split and Class Imbalance

An 80:20 train–test partition was generated using stratification on `Diabetes_012`, with random seed 42. Stratification preserves the approximate class proportions in both subsets, which is important because prediabetes accounts for only 1.8% of the observations. Without this constraint, the test partition might contain too few class 1 cases for a meaningful assessment. The resulting training and test sets contain 202,944 and 50,736 records, respectively; the test-set class counts are 42,741, 926, and 7,069.

All baseline classifiers were fitted without class weighting, allowing the effect of the original class distribution to be observed. We additionally evaluated a Random Forest configured with `class_weight="balanced_subsample"`. For each bootstrap sample, this option assigns class k a weight of $N/(3N_k)$; the approximate weights on the complete dataset are 0.40, 18.3, and 2.4. Other imbalance strategies, including SMOTE [14] and decision-threshold adjustment, are reserved for subsequent experiments (Table VIII).

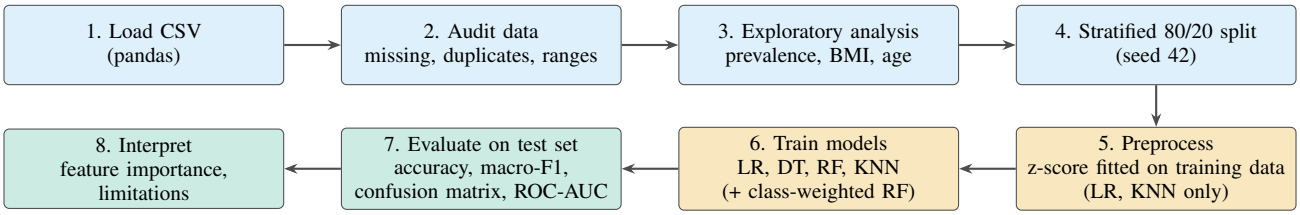


Fig. 1. Workflow followed in this study, from raw CSV to interpretation.

TABLE III
MODEL CONFIGURATIONS (ALL WITH `RANDOM_STATE`=42)

Model	Main settings	Scaling
Logistic Reg.	multinomial, L2, C=1, max 1000 iterations	z-score
Decision Tree	Gini, max depth 10	none
Random Forest	100 trees, Gini, unlimited depth	none
KNN	k=5, Euclidean, uniform weights	z-score
RF (weighted)	as RF, balanced_subsample, min leaf size 5	none

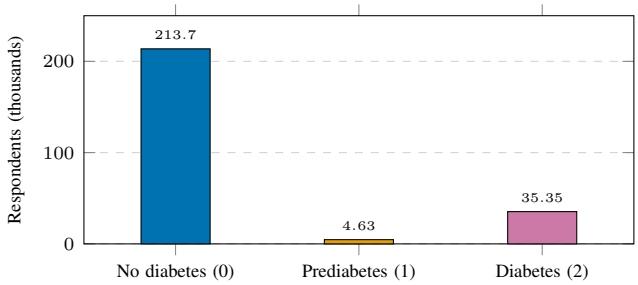


Fig. 2. Distribution of the target variable `Diabetes_012` (253,680 respondents; 84.2%, 1.8%, and 13.9%).

D. Settings and Reproducibility

The configurations used in the experiments are detailed in Table III. No systematic hyperparameter search was performed; standard settings were retained, with a maximum depth of 10 imposed on the Decision Tree. The implementation environment consisted of Python 3.13, scikit-learn 1.9.1, pandas 3.0.5, and NumPy 2.5.3. The principal results can be regenerated from the CSV using `analyze.py`; a supplementary script performs the additional analyses and the duplicate-free evaluation.

V. EXPLORATORY DATA ANALYSIS

A. Target Distribution

The class frequencies plotted in Fig. 2 reveal a pronounced imbalance. A trivial classifier that assigns every respondent to class 0 would achieve 84.24% accuracy, including on the test partition (42,741 of 50,736 cases), despite failing to identify any respondent with prediabetes or diabetes. Its macro-F1 is approximately 0.305, derived from an F1 of 0.914 for class 0 and zero for the remaining classes. This majority-class predictor is used as the reference point for the model comparisons.

B. Health-Indicator Prevalence

Figure 3 presents the within-class prevalence of eight binary health indicators. High blood pressure increases across the outcome groups, from 37.1% in class 0 to 62.9% in class 1 and 75.3% in class 2. A similar pattern appears for high cholesterol, with corresponding proportions of 37.9%, 62.1%, and 67.0%. The reported prevalence of heart disease or heart attack is 22.3% among respondents with diabetes compared with 7.2% in class 0. Difficulty walking rises from 13.2% to 37.1%, while stroke increases from 3.2% to 9.3%. Physical activity shows the reverse pattern, declining from 77.9% to 63.1% across these groups. Heavy alcohol consumption is less frequently reported in class 2 (2.4%) than in class 0 (6.2%); this association should not be interpreted as evidence of a protective effect, since behaviour may change after diagnosis. Across several indicators, the prediabetes group resembles the diabetes group more closely than the non-diabetes group, which may contribute to the difficulty of separating the two positive classes.

The combination of the two conditions is also informative. Diabetes is reported by 4.2% of respondents with neither high blood pressure nor high cholesterol (101,920 individuals). The corresponding proportion is 10.4% for high cholesterol alone (42,931 individuals), 16.7% for high blood pressure alone (44,169 individuals), and 29.7% when both conditions are reported (64,660 individuals).

C. Duplicates and Data Quality

After exact duplicates are excluded, 229,781 observations remain. Table IV shows that 23,648 of the removed records (98.9%) belong to class 0; only two are from class 1 and 249 from class 2. This removal changes the class proportions modestly, as illustrated in Fig. 4: the share of class 0 decreases from 84.2% to 82.7%, whereas the proportions of classes 1 and 2 increase to 2.0% and 15.3%, respectively. The imbalance nevertheless remains substantial. No other data-quality issue was identified in the audit. BMI values range from 12 to 98, with a mean of 28.4 and standard deviation of 6.6. Although the endpoints are unusual, they were retained because they may reflect extreme self-reported values.

D. BMI

Mean BMI differs among the three outcome categories: 27.7 in class 0, 30.7 in class 1, and 31.9 in class 2; the respective

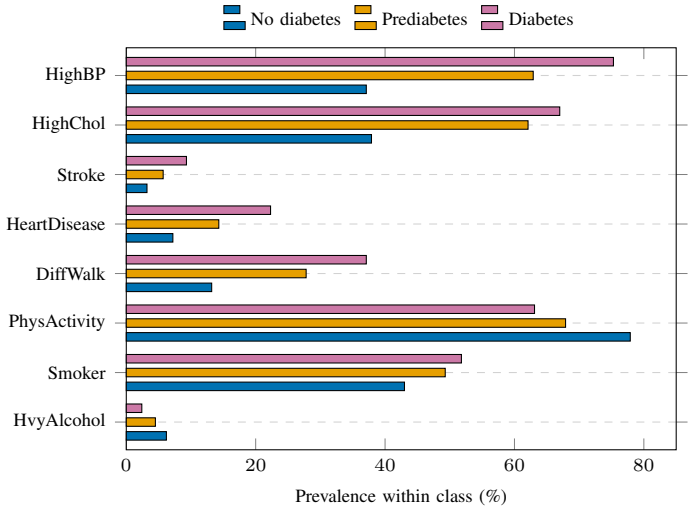


Fig. 3. Prevalence of selected binary health indicators in each target class. Heart disease is "HeartDiseaseorAttack"; heavy alcohol is "HvyAlcoholConsump".

TABLE IV
EFFECT OF REMOVING DUPLICATE ROWS BY CLASS

Class	Original	Deduplicated	Removed
0 (no diabetes)	213,703	190,055	23,648
1 (prediabetes)	4,631	4,629	2
2 (diabetes)	35,346	35,097	249
Total	253,680	229,781	23,899

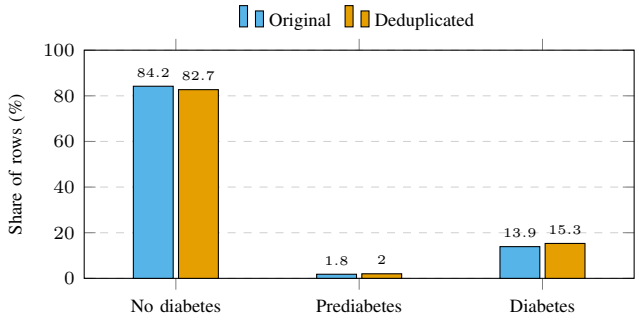


Fig. 4. Class shares before (253,680 rows) and after (229,781 rows) removing duplicate rows.

medians are 27, 30, and 31. The category-level results in Fig. 5 show a rising proportion of diabetes with increasing BMI. The reported diabetes share is 5.7% for the normal-weight interval (18.5–24.9), 11.4% for the overweight interval, 19.2% for BMI 30–34.9, and 29.9% for BMI values of at least 35. The last figure is more than five times the normal-weight proportion. Prediabetes also increases, from 1.0% to 3.1%. The underweight category contains only 3,127 respondents, making its 5.4% estimate less stable. Because these observations are cross-sectional, the patterns indicate association rather than a causal effect of body weight.

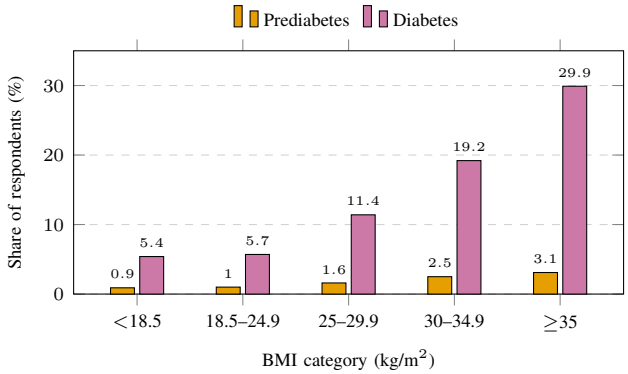


Fig. 5. Percentage of respondents in each BMI category who report prediabetes or diabetes (group sizes: 3,127; 68,953; 93,749; 53,451; 34,400).

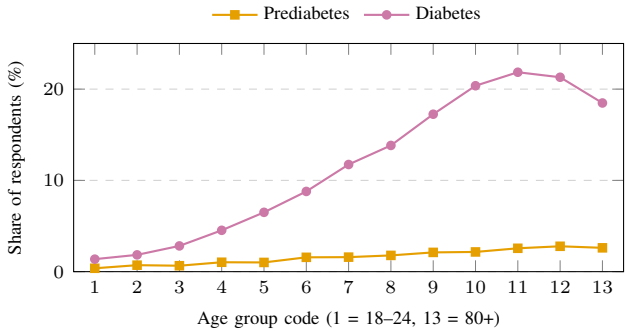


Fig. 6. Percentage of respondents with prediabetes or diabetes by BRFSS age group.

E. Age

The survey represents age through 13 ordered bands, each spanning approximately five years. As shown in Fig. 6, the diabetes proportion increases from 1.4% in the 18–24 group to 21.9% among those aged 70–74 (code 11), followed by 21.3% in the 75–79 group. It is 18.5% among respondents aged 80 or above. The prediabetes proportion changes more gradually, increasing from 0.4% to 2.8%. Mean age codes are 7.8, 9.1, and 9.4 for classes 0, 1, and 2, respectively. The lower value in the oldest group may be related to survival patterns or differences in survey participation; it should not be interpreted as evidence of reduced risk at older ages.

F. General Health, Income, and Education

Among the examined variables, self-assessed general health has the largest observed gradient. Figure 7a shows that diabetes prevalence increases from 2.5% for respondents selecting "excellent" health to 37.9% for those selecting "poor" health. Differences are also visible across socioeconomic categories (Fig. 7b). The diabetes proportion decreases from 24.3% in the lowest income category to 8.0% in the highest. Across education levels, it declines from approximately 29% at the lower end to 9.7% among college graduates. The two lowest education categories include only 174 and 4,043 respondents, respectively, so these estimates should be treated cautiously.

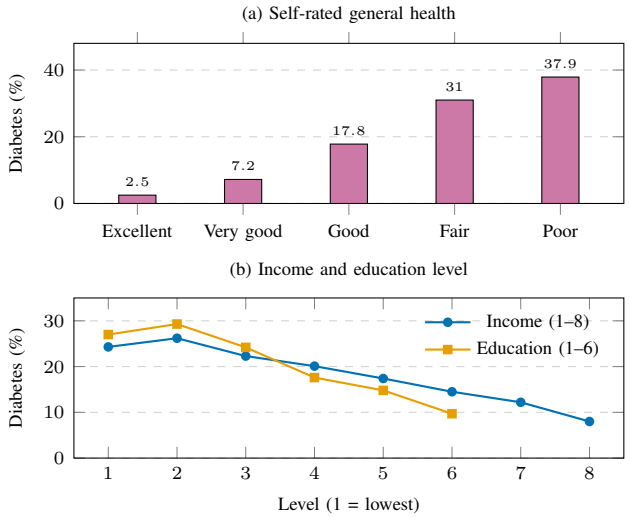


Fig. 7. Share of respondents with diabetes (class 2) by (a) self-rated general health and (b) income and education level.

Pearson correlations with the ordinal outcome are limited in magnitude: GenHlth (0.30), HighBP (0.27), BMI (0.22), DiffWalk (0.22), HighChol (0.21), and Age (0.19) are positive, whereas Income (-0.17), Education (-0.13), and PhysActivity (-0.12) are negative. The difference by Sex is comparatively small: 15.2% for code 1 and 13.0% for code 0. These associations suggest that the available predictors provide only moderate separation between outcome groups.

VI. Algorithms

The following summaries describe the four classifiers used in the experiments; their established formulations are documented in the cited references.

Logistic Regression [9] estimates a score for each outcome class and converts these scores into a probability distribution using the softmax expression:

$$P(y{=}k \mid \mathbf{x}) = \frac{e^{\mathbf{w}_k^\top \mathbf{x}+b_k}}{\sum_{j=0}^{2} e^{\mathbf{w}_j^\top \mathbf{x}+b_j}}, \tag{1}$$

The model parameters are estimated by minimising the cross-entropy objective with an L2 regularisation term:

$$\min_{W,\mathbf{b}} \ \tfrac{1}{2}\|W\|_2^2 + C\sum_{i=1}^{N} -\log P(y_i \mid \mathbf{x}_i), \quad C=1. \tag{2}$$

Consequently, the resulting class boundaries are linear with respect to the input attributes.

A **Decision Tree** [10] constructs a sequence of feature-based partitions. At each node, it selects a split that produces the largest reduction in Gini impurity:

$$G(t) = 1 - \sum_{k=0}^{2} p_{tk}^2, \quad \Delta G = G(t) - \tfrac{N_L}{N_t}G(t_L) - \tfrac{N_R}{N_t}G(t_R), \tag{3}$$

Here, p_{tk} denotes the fraction of observations from class k at node t, while N_L and N_R represent the numbers of observations in its left and right child nodes. A terminal node assigns the class with the largest count. The maximum tree depth in this experiment is 10.

Algorithm 1 Evaluation Procedure

Require: CSV file, model set $\mathcal{M}$, seed $s = 42$

Load data; check rows, columns, missing cells, duplicates
$(X_{tr}, y_{tr}), (X_{te}, y_{te}) \leftarrow$ stratified 80/20 split on y
Fit scaler on X_{tr}; scale X_{tr} and X_{te} for LR and KNN
for each model $m \in \mathcal{M}$ **do**
    Fit m on the training data
    $\hat{y} \leftarrow m.\text{predict}(X_{te})$; $\hat{P} \leftarrow m.\text{predict_proba}(X_{te})$
    Record accuracy, macro P/R/F1, per-class recall, confusion matrix, macro OvR AUC
end for
Repeat on the deduplicated data; extract RF importances and LR coefficients

The **Random Forest** classifier [11] combines predictions from an ensemble of $B = 100$ decision trees. Each tree is fitted to a bootstrap sample, and a randomly selected subset of predictors is considered at each split. The final class is selected from the averaged class probabilities:

$$\hat{y} = \arg\max_k \ \frac{1}{B}\sum_{b=1}^{B} \hat{P}_b(y{=}k \mid \mathbf{x}). \tag{4}$$

The forest also provides impurity-based feature-importance scores. Such scores can favour predictors with a larger number of distinct values, as discussed in [18].

For **k-Nearest Neighbours** (KNN) [12], the predicted label is determined by the majority class among the five training observations closest to the test respondent:

$$d(\mathbf{x},\mathbf{x}') = \sqrt{\textstyle\sum_{j=1}^{21}(x_j - x'_j)^2}, \quad \hat{y} = \arg\max_k \sum_{i\in\mathcal{N}_5(\mathbf{x})} \mathbf{1}[y_i = k]. \tag{5}$$

Because many predictors are binary, different observations can have identical or very similar distances. A respondent from a minority class may therefore have a neighbourhood dominated by class 0, affecting the assigned label.

VII. Evaluation Metrics

Let C_{ij} denote the number of test observations whose actual class is i and whose predicted class is j. For each class k, TP_k, FP_k, and FN_k indicate its true-positive, false-positive, and false-negative counts. The evaluation measures are defined as follows:

$$\text{Accuracy} = \frac{\sum_k C_{kk}}{N}, \tag{6}$$

$$P_k = \frac{TP_k}{TP_k + FP_k}, \quad R_k = \frac{TP_k}{TP_k + FN_k}, \tag{7}$$

$$F1_k = \frac{2P_kR_k}{P_k + R_k}, \quad \text{Macro-F1} = \frac{1}{3}\sum_{k=0}^{2} F1_k. \tag{8}$$

Each observation contributes equally to accuracy; given the class distribution, this measure is therefore strongly influenced

TABLE V
TEST-SET PERFORMANCE (SINGLE SPLIT, SEED 42)

Model	**Acc.** (%)	**Prec.** (macro)	**Rec.** (macro)	**F1** (macro)	**AUC** (OvR)
Logistic Reg.	84.55	0.459	0.383	0.392	**0.781**
Decision Tree	84.53	**0.525**	0.391	0.402	0.758
Random Forest	84.26	0.452	0.388	0.398	0.742
KNN (k=5)	83.25	0.457	0.388	0.396	0.656
RF, class-weighted	76.94	0.433	**0.492**	**0.443**	0.763
Majority baseline	84.24	—	0.333	0.305	0.500

by class 0. Recall is particularly relevant to screening because it quantifies the proportion of actual cases detected by a classifier. Macro-F1 gives the three classes equal influence by averaging their individual F1 scores, meaning that poor performance on the relatively rare prediabetes group is not hidden by strong results on the majority group [15]. Macro precision and macro recall are computed as the unweighted means of the corresponding class-wise values.

Multiclass ROC-AUC is calculated using a one-versus-rest procedure. In turn, each outcome category is treated as the positive class, and the model probability for that category is used to rank all test observations. The area under each ROC curve [16] is then averaged across the three categories:

$$\mathrm{AUC}_{\mathrm{OvR}} = \frac{1}{3}\sum_{k=0}^{2} \mathrm{AUC}_k. \tag{9}$$

Unlike accuracy, recall, and F1, AUC is not tied to a single classification threshold. It evaluates the quality of the model's probability ranking rather than only its default class assignments. For this reason, AUC is reported alongside the threshold-dependent measures. Hand and Till describe an alternative based on pairwise class comparisons [17].

VIII. RESULTS AND DISCUSSION

The results below are based on a single stratified test partition containing 50,736 respondents, generated with seed 42. Confidence intervals were not estimated.

A. Overall Comparison

Table V and Fig. 8 summarise the test-set comparison. Among the unweighted classifiers, Logistic Regression records the highest accuracy (84.55%) and macro AUC (0.781), while the Decision Tree obtains the largest macro-F1 value (0.402). Differences between the conventional models are small. Logistic Regression and the Decision Tree exceed the 84.24% majority baseline by approximately 0.3 percentage points; Random Forest reaches 84.26%, almost identical to the baseline, and KNN records 83.25%. Thus, under the selected default configurations, none of the four standard classifiers provides a substantial accuracy improvement over always predicting the majority class.

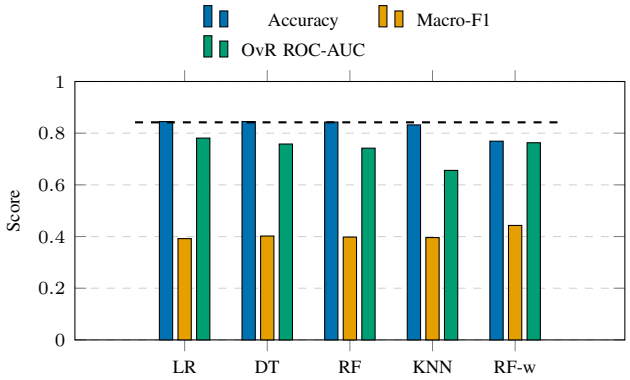


Fig. 8. Model comparison on the test set. RF-w is the class-weighted Random Forest; the dashed line marks the majority-class accuracy (0.842).

TABLE VI
PER-CLASS RECALL ON THE TEST SET

Model	**Class 0**	**Class 1**	**Class 2**
Logistic Regression	0.975	0.000	0.174
Decision Tree	0.971	0.003	0.199
Random Forest	0.968	0.000	0.197
KNN (k=5)	0.954	0.003	0.207
RF, class-weighted	0.803	0.006	0.666

B. Where the Errors Are

The class-wise values in Table VI expose weaknesses that are not apparent from overall accuracy. The four unweighted classifiers correctly identify 95–97% of class 0, but detect only 17.4–20.7% of respondents in class 2. Their recall for prediabetes is effectively zero, ranging from 0 to 0.3%; even the best-performing models identify only three of the 926 test cases in this category. Logistic Regression does not assign any observation to class 1. The left confusion matrix in Fig. 9 illustrates the same issue for Random Forest: 5,669 of the 7,069 diabetic respondents are classified as having no diabetes.

Introducing class weights produces a different error profile, as shown in the right panel of Fig. 9. The weighted forest correctly detects 4,707 of the 7,069 class 2 respondents, increasing diabetes recall to 66.6% from 19.7%. Its macro-F1 also improves, from 0.398 to 0.443. This gain is accompanied by lower overall accuracy (76.9%), because 8,212 class 0 respondents, or 19.2% of that group, are incorrectly flagged. Class 1 recall remains low at 0.6% (6 of 926 cases). The suitability of this trade-off depends on the relative consequences of additional clinical tests and missed cases, which requires clinical and operational judgement.

C. Effect of Removing Duplicates

To assess sensitivity to repeated observations, we reran the analysis after deduplication. This produced 183,824 training records and 45,957 test records; the test classes contain 38,012, 926, and 7,019 observations. The majority-class reference accuracy falls to 82.71%. Results are listed in Table VII. Macro-F1 remains similar, ranging from 0.388 to 0.391 for

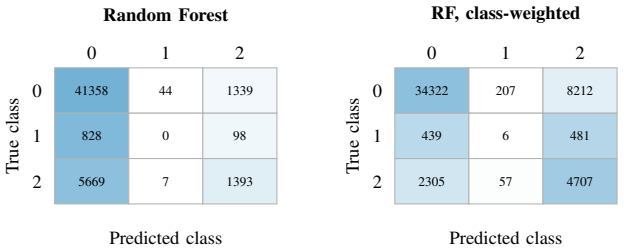


Fig. 9. Confusion matrices on the test set (counts; shading shows the row-normalised share of each true class). Left: Random Forest. Right: class-weighted Random Forest.

TABLE VII
RESULTS AFTER REMOVING DUPLICATE ROWS (TEST SET OF 45,957)

Model	Acc. (%)	Macro-F1	AUC	Recall, class 2
Logistic Reg.	83.30	0.390	0.769	0.173
Decision Tree	83.23	0.388	0.748	0.169
Random Forest	82.40	0.391	0.718	0.192
KNN (k=5)	81.52	0.391	0.648	0.206
RF, class-weighted	75.98	0.448	0.749	0.653
Majority baseline	82.71	0.302	0.500	0.000

the standard models and reaching 0.448 for the weighted forest. Diabetes recall is still between 17% and 21% for the unweighted classifiers, compared with 65.3% for the weighted model. AUC decreases for all classifiers, with the largest changes observed for Random Forest (0.742 to 0.718) and Logistic Regression (0.781 to 0.769). Although duplicated records may have influenced ranking performance to some extent, their removal does not alter the central conclusions.

D. Reading the ROC-AUC Values

The highest macro one-vs-rest AUC is obtained by Logistic Regression (0.781), followed by the class-weighted forest (0.763), Decision Tree (0.758), standard Random Forest (0.742), and KNN (0.656). In this experiment, the linear classifier provides stronger overall risk ranking than the more flexible alternatives. One possible explanation is that the information contained in the 21 survey variables is captured reasonably well by additive relationships; however, the present analysis cannot establish this explanation. Only the macro-averaged AUC is reported. Class-specific ROC curves were not generated, so performance for prediabetes cannot be directly compared with performance for diabetes.

E. Which Features Matter

The ten largest impurity-based Random Forest importance values are plotted in Fig. 10. BMI has the highest value (0.183), followed by Age (0.123), Income (0.098), PhysHlth (0.084), Education (0.071), GenHlth (0.070), and MentHlth (0.065). These values require cautious interpretation because impurity-based importance can favour variables with many possible values [18]; consequently, binary indicators such as HighBP (0.042) may receive comparatively modest scores. The absolute standardised Logistic Regression coefficients, averaged across classes, produce a different ordering: GenHlth (0.21), Age (0.17), BMI (0.17), HighBP (0.13), and HighChol (0.13). Both approaches highlight BMI and age and assign importance to socioeconomic attributes, but they differ in the emphasis placed on health-status variables. These rankings are descriptive indicators of model behaviour, not evidence of causation.

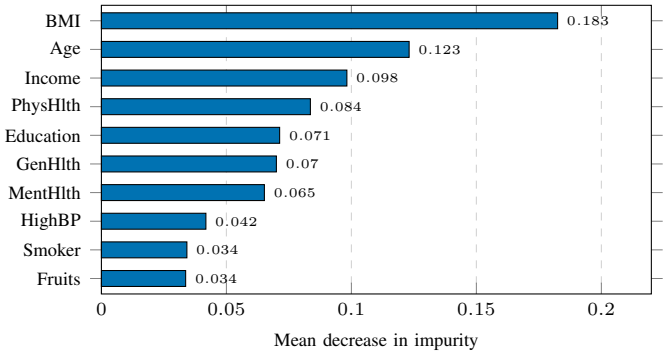


Fig. 10. Top-10 Random Forest feature importances (100 trees, unweighted).

F. Discussion

The findings support three observations. First, the survey attributes provide limited discriminatory information: relative to the majority-class reference, the standard classifiers offer little practical improvement, consistent with the modest associations reported in Section V. Second, the treatment of class imbalance has a greater effect on diabetes detection than the choice among the four conventional algorithms. Class weighting increases diabetes recall by nearly 47 percentage points, whereas the unweighted models differ by fewer than four points. Third, identifying prediabetes remains particularly difficult. This category is small and shares several indicator patterns with the diabetes group, while its self-reported label may also be affected by awareness. Any future screening application would need a decision threshold selected according to the consequences of missed cases and false alarms. Its output should communicate estimated risk and must not be presented as a medical diagnosis.

IX. LIMITATIONS

- **Self-reported cross-sectional observations.** The predictors and labels are obtained through a telephone survey and may be affected by recall or awareness bias. Respondents with undiagnosed diabetes may appear in class 0, and behavioural changes following diagnosis can complicate observed associations.
- **Single train–test partition.** The reported comparison uses one random seed and one split. Without cross-validation or confidence intervals, small differences, such as the macro-F1 gap between the tree and forest, cannot be considered conclusive.
- **Limited parameter optimisation.** Most hyperparameters use standard settings, and the results therefore do not represent the best achievable performance of each method.

TABLE VIII
EXPERIMENTS NOT YET RUN (RESULTS PENDING)

Planned experiment	Metric	Result
5-fold stratified CV with 95% CI	macro-F1, AUC	[pending]
Hyperparameter search (all models)	macro-F1	[pending]
SMOTE / threshold tuning	recall (class 1, 2)	[pending]
Subgroup (sex, age, income) analysis	recall, FPR	[pending]

Ordered categorical fields are also treated as numeric inputs.

- **Restricted imbalance strategies.** The experiments examine class weighting only. Synthetic oversampling such as SMOTE [14], threshold selection, and boosted-tree methods remain unevaluated.
- **Population coverage and fairness.** The dataset represents U.S. adults surveyed in 2015. Performance was not separately assessed across age, income, or sex groups, leaving subgroup reliability and fairness unresolved [19]. None of the evaluated models has clinical validation, and none should be used to diagnose diabetes.

X. FUTURE WORK

Future work will begin with the outstanding evaluations listed in Table VIII, including stratified cross-validation, parameter optimisation, and additional imbalance-handling methods. Further experiments may compare gradient-boosted classifiers and probability-calibration techniques. We will also consider combining prediabetes and diabetes into a binary outcome, or modelling the ordered nature of the three labels directly. Decision thresholds should be selected on validation data using explicit costs for false negatives and false positives. In addition, SHAP-based explanations could complement impurity-based importance, and a Streamlit interface may be developed to display estimated risk from user-entered attributes together with an explicit medical disclaimer.

XI. CONCLUSION

The 2015 BRFSS-derived diabetes indicator dataset was examined to assess its quality, explore relationships between survey attributes and diabetes status, and compare four machine-learning classifiers. The file contains 253,680 records and 22 columns, has no missing entries, and includes 23,899 duplicate rows. The outcome distribution is 84.2% no diabetes, 1.8% prediabetes, and 13.9% diabetes. Observed diabetes prevalence increases from 5.7% in the normal-BMI group to 29.9% among respondents with BMI of at least 35, and from 2.5% to 37.9% across the "excellent" to "poor" general-health categories. Despite these associations, Logistic Regression, Decision Tree, Random Forest, and KNN achieve 83.2–84.6% accuracy, close to the 84.2% majority baseline. Their macro-F1 values remain near 0.39–0.40, and prediabetes detection is minimal. Removing duplicate records does not materially alter this pattern. Logistic Regression gives the strongest risk ranking, with an AUC of 0.781. Class weighting improves the Random Forest macro-F1 to 0.443 and diabetes recall to 66.6%, while reducing accuracy by 7.3 percentage points. The results underline the need to prioritise recall and macro-F1 over accuracy for this imbalanced problem. Cross-validation, model tuning, and subgroup evaluation are necessary before considering practical deployment.

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